Large Language Model-based Chatbot as a Source of Advice on First Aid in Heart Attack

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Abstract: The ability of the cutting-edge large language model-powered chatbots to generate human-like answers to user questions hypothetically could be utilized for providing real-time advice on first aid for witnesses of cardiovascular emergencies. This study aimed to evaluate quality of the chatbot responses to inquiries on help in heart attack. The study simulated interrogation of the new Bing chatbot (Microsoft Corporation, USA) with the "heart attack what to do" prompt coming from 3 countries, the Gambia, India and the USA. The chatbot responses (20 per country) were evaluated for congruence with the International First Aid, Resuscitation, and Education Guidelines 2020 using a checklist. For all user inquiries, the chatbot provided answers containing some guidance on first aid. However, the responses commonly left out some potentially life-saving instructions, for instance to encourage the person to stop physical activity, to take antianginal medication, or to start cardiopulmonary resuscitation for unresponsive abnormally breathing person. Mean percentage of the responses

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having full congruence with the checklist criteria varied from 7.3 for India to 16.8 for the USA. A quarter of responses for the Gambia and the USA, and 45.0% for India contained superfluous guidelines-inconsistent directives. The chatbot advice on help in heart attack has omissions, inaccuracies and misleading instructions, and therefore the chatbot cannot be recommended as a credible source of information on first aid. Active research and organizational efforts are needed to mitigate the risk of uncontrolled misinformation and establish measures for guaranteeing trustworthiness of the chatbot-mediated counseling. (Curr Probl Cardiol 2024;49:102048.)

Introduction

Application of innovative information and communication technologies has been identified by the international cardiologic community as a promising way to reduce morbidity and mortality from cardiovascular diseases worldwide.[1-3] A range of studies has demonstrated the benefits of digital interventions in different domains of cardiovascular health, including patient lifestyle behavior change,[4] noninvasive telemonitoring,[5] cardiovascular disease self-management,[6] and telerehabilitation.[7] The growing availability of internet and personal mobile devices, along with the shortages of medical personnel, and increased life expectancy with associated impeded mobility of the aging global population suggest that utilization of digital health solutions would continuously rise in the near future.[1] As such, it is important to promote scientific research aimed at investigating the value and possible impact of emerging digital technologies on population health, particularly in the field of cardiovascular care.

The release of an openly available artificial intelligence (AI)-powered conversational tool called the Chat Generative Pretrained Transformer (ChatGPT; OpenAI, San Francisco, California, USA) in late 2022 has garnered extraordinary public attention to the large language model (LLM) chatbots. Concurrent with considerable hype in media associated with the outstanding ability of these chatbots to perform a range of natural language processing tasks, including machine translation, sentiment analysis, text summarisation and human-like question answering,[8] the LLM technologies have rapidly entered the mainstream of contemporary health research. The ever-growing body of publications suggests the

potential utility of the chatbots in healthcare education, biomedical data analysis, and clinical practice,[9] simultaneously raising some concerns regarding the trustworthiness of the chatbot-generated information and possibility of intentional misuse of the LLM tools.[10,11]

Hypothetically, publicly available AI chatbots can be employed as interactive question-answering systems meant for providing real-time informational support for patients or witnesses of health emergencies.[12-14] Similar to dispatcher prearrival instructions communicated over the telephone, the chatbot-generated advice received through a handheld mobile device or a smart speaker could contribute to early recognition of a life-threatening condition, rapid activation of emergency medical services (EMS) and timely first aid intervention. Such a solution, on the premise of guaranteed quality, could be especially valuable as a means of support in cardiovascular emergencies for remote and underserved populations, for instance in low- and middle-income countries where the rates of internet and mobile technologies' penetration[15] outrun development of EMS[16] and coverage of the community with high-quality first aid training,[17] but the incidence and mortality from cardiovascular diseases remains high.[18]

This study set out to evaluate quality of the LLM-powered chatbot responses to user inquiries on help in heart attack in terms of congruence with current guidelines on first aid. A secondary objective was to determine whether there are differences in the chatbot performance when the information search is conducted for different countries.

Material and Methods

The new Bing Chatbot

The new Bing was released by Microsoft Corporation (Redmond, Washington, USA) in February 2023 as an AI-enhanced web search "copilot" with chatbot functionality. The new Bing reacts to user inquiries by performing internet search, ranking content found online with consideration of its relevance, quality, credibility, freshness, and other characteristics, and providing a summarized "grounded" response based on the top web search results, accompanied with links to the source websites.[19,20] The summarized responses are created by the Generative Pretrained Transformer 4 (GPT-4) LLM developed by OpenAI (San Francisco, California, USA).[21] This AI model is trained on a vast dataset of human-generated text to probabilistically predict and repeatedly generate the next word in a sentence based on the previous words, thereby

producing coherent and natural-sounding text.[8] By means of the chatbot functionality, users of the new Bing can perform web search conversationally, in particular ask follow-up questions, request clarification or additional information to initial search to get personalized replies.[19]

Study Design

The study was designed with consideration of prior related research.[13,22]

The study simulated the chatbot interrogation originating from 3 countries having English as the official language, with different income levels: the Gambia (low-income country with the highest internet penetration [percentage of individuals using the Internet] among the low-income countries using English as the official language – 32% as of 2020[15]), India (middle-income country; internet penetration – 43%) and the USA (high-income country; internet penetration – 91%).

To select the search query, online search popularity of the keyword combinations ("chest pain" OR "heart attack") AND ("first aid" OR "help" OR "how" OR "what") was investigated with Google Trends (Google LLC, Mountain View, California, USA) for past 5 years worldwide and by the countries. Consequently, "heart attack what to do" was chosen as the most popular query.

In May 2023, the new Bing chatbot was repeatedly inquired, "heart attack what to do" using Microsoft Edge web browser (Microsoft Corporation, Redmond, Washington, USA) running on a personal computer. In the chatbot settings, the search locations were set to capitals of the countries under study, namely Banjul for the Gambia, Delhi for India, and Washington DC for the US; and "More Precise" (concise and straightforward) chatbot's conversation style was selected. The chatbot was prompted 20 times per country. Before each chatbot inquiry, all search, browsing and download history, cache, and cookies were deleted from the web browser and the Microsoft account.

The chatbot textual responses were tabulated and subsequently independently evaluated by the authors using the checklist developed based on the provisions of the International Federation of Red Cross and Red Crescent Societies' International First Aid, Resuscitation, and Education Guidelines 2020.[23] During the evaluation, the responses' content was judged in terms of satisfaction of each checklist item wording as True (completely satisfied), Partially True (satisfied in part), or Not True (not satisfied/absent). After comparing and discussing the results, consensus rating was achieved. Additionally, when the chatbot responses contained

any excessive advice noncompliant with the guidelines, this was recorded. Length (number of sentences) and readability (Flesch-Kincaid Grade Level[24]) of the chatbot responses were determined using the open online tool Datayze.[25]

Content of the source web articles referenced within the chatbot responses was evaluated for compliance with the checklist criteria following the same procedure with the focus on the main articles' text (i.e. omitting other webpage sections or links to external resources). Respective source webpages were classified as Academic, Charity/Nonprofit entity, Commercial, Government, and Social media.

The obtained data were submitted to descriptive statistical analysis, Mann−Whitney U test, and Wilcoxon signed-rank test.

All original data generated in this research are publicly available in online repository as a dataset.[26]

Results

The chatbot replied to all user inquiries by providing the query-related textual answers containing some advice on how to give help to a person with heart attack.

For the studied countries, the chatbot responses had similar length, whereas the readability differed, being lower for the Gambia and the USA (requiring approximately 12th education grade to understand the text) than for India (requiring approximately 10th grade; $P \leq 0.008$; Table 1).

The checklist-based evaluation of the chatbot responses' content revealed poor coverage of the guidelines-concordant instructions on first aid in heart attack (Table 2 and Dataset[26]). Except in a few cases, the chatbot advice omitted recommendations to get the person into a comfortable position, encourage them to stop physical activity, reassure them, monitor their responsiveness and breathing, and to ask a bystander to get an automated external defibrillator (AED) when possible if EMS arrival is delayed. A portion of responses, variable depending on the country,

Table 1. The chatbot responses' length and readability

Parameters	The Gambia ($N = 20$)	India ($N = 20$)	The USA ($N = 20$)
Number of sentences, median [IQR]	8.0 (7.0-9.0)	8.0 (6.0-9.0)	8.0 (8.0-9.0)
Readability, Flesch-Kincaid grade level, median [IQR]	12.3 (11.4-12.9)	9.6 (7.6-11.9)	12.1 (10.1-12.6)

Abbreviations: IQR, interquartile range.

Table 2. Congruence of the chatbot responses' content with first aid guidelines as judged by satisfaction of the checklist criteria

Checklist criteria	Completely satisfied			Partially satisfied			Completely or partially satisfied		
	GM (*N* = 20), % (*n*)	IN (*N* = 20), % (*n*)	US (*N* = 20), % (*n*)	GM (*N* = 20), % (*n*)	IN (*N* = 20), % (*n*)	US (*N* = 20), % (*n*)	GM (*N* = 20), % (*n*)	IN (*N* = 20), % (*n*)	US (*N* = 20), % (*n*)
1. Does the response instruct to help the person to get into a comfortable position?	0.0 (0)	0.0 (0)	0.0 (0)	10.0 (2)	0.0 (0)	0.0 (0)	10.0 (2)	0.0 (0)	0.0 (0)
2. Does the response instruct to encourage the person to refrain from physical activity?	0.0 (0)	0.0 (0)	0.0 (0)	5.0 (1)	0.0 (0)	0.0 (0)	5.0 (1)	0.0 (0)	0.0 (0)
3. Does the response instruct to help the person to take their antianginal medication?	90.0 (18)	60.0 (12)	85.0 (17)	0.0 (0)	0.0 (0)	0.0 (0)	90.0 (18)	60.0 (12)	85.0 (17)
4. Does the response instruct to access EMS immediately?	5.0 (1)	20.0 (4)	100.0 (20)	95.0 (19)	80.0 (16)	0.0 (0)	100.0 (20)	100.0 (20)	100.0 (20)
5. Does the response instruct to ask a bystander to bring an AED if possible if EMS is delayed?	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)
6. Does the response instruct to suggest the person to consider chewing a dose of acetylsalicylic acid (150-300 mg)?	0.0 (0)	0.0 (0)	0.0 (0)	100.0 (20)	100.0 (20)	100.0 (20)	100.0 (20)	100.0 (20)	100.0 (20)
7. Does the response instruct not to give acetylsalicylic acid to the person if they are allergic to it, have a bleeding disorder, or have already taken the recommended dose?	0.0 (0)	0.0 (0)	0.0 (0)	100.0 (20)	70.0 (14)	100.0 (20)	100.0 (20)	70.0 (14)	100.0 (20)

(*continued on next page*)

Table 2. (*continued*)

Checklist criteria	Completely satisfied			Partially satisfied			Completely or partially satisfied		
	GM (*N* = 20), % (*n*)	IN (*N* = 20), % (*n*)	US (*N* = 20), % (*n*)	GM (*N* = 20), % (*n*)	IN (*N* = 20), % (*n*)	US (*N* = 20), % (*n*)	GM (*N* = 20), % (*n*)	IN (*N* = 20), % (*n*)	US (*N* = 20), % (*n*)
8. Does the response instruct to reassure the person?	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)
9. Does the response instruct to monitor person's breathing and responsiveness?	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)
10. Does the response instruct to start CPR if the person becomes unresponsive with abnormal breathing?	0.0 (0)	0.0 (0)	0.0 (0)	90.0 (18)	55.0 (11)	85.0 (17)	90.0 (18)	55.0 (11)	85.0 (17)
11. Does the response instruct to use an AED if available if the person becomes unresponsive with abnormal breathing?	0.0 (0)	0.0 (0)	0.0 (0)	90.0 (18)	25.0 (5)	65.0 (13)	90.0 (18)	25.0 (5)	65.0 (13)
Mean percentage	8.6	7.3	16.8	44.5	30.0[a]	31.8[a]	53.2	37.3[a]	48.6[a]

[a]Significant difference vs. GM, $P<0.05$.

Abbreviations: AED, automated external defibrillator; CPR, cardiopulmonary resuscitation; EMS, emergency medical services; GM, the Gambia; IN, India; US, the United States.

left out instructions to help the person to take their antianginal medication, and to start cardiopulmonary resuscitation (CPR), or use an AED if available, when the person presents unresponsive with abnormal breathing.

When the chatbot responses included relevant advice, the instructions were commonly judged as partially compliant with the guidelines because of inaccuracies or omissions of important details. In particular, whereas the responses always suggested to prompt the person to chew a dose of acetylsalicylic acid, they never mentioned the recommended dose of the drug (150-300 mg[23]). As concerns the instruction to activate EMS, no response for the Gambia or India recommended to call the country-specific national emergency telephone numbers, but on the contrary, in most cases the chatbot mistakenly advised to call the US emergency number 911. Further, the chatbot always omitted abnormal breathing (gasping) among the criteria for starting CPR, and commonly suggested to initiate CPR when the person has no pulse, whereas the pulse check is not recommended for recognition of cardiac arrest by lay rescuers.[27]

The between-country comparison showed that the chatbot responses for the Gambia had higher compliance of the first aid advice with the guidelines, than those for India or the USA (see Table 2).

Additionally, 25.0% (n = 5) responses for the Gambia, 45.0% (n = 9) for India, and 25.0% (n = 5) for the USA contained some guidelines-inconsistent superfluous directives (see Dataset[26]), including the recommendation to loosen person's clothes, open windows, move the patient to a more ventilated area, make the patient sit or provide them a reclined position, check patient's pulse (as part of recognition of cardiac arrest), start CPR in unconscious person (without mention of other essential criteria for recognition of cardiac arrest, i.e. absent or abnormal breathing). Several responses admitted the possibility for the patient to drive themselves to a hospital when there is no other option to get help.

The number of source web articles cited in a single chatbot response varied from 1 to 3 for the Gambia (median [IQR]: 2.0 [1.0-2.0]), 1-5 for India (1.0 [1.0-1.0]), and 1-3 for the USA (2.0 [1.0-2.0]; see Dataset[26]). Table 3 shows distribution of the source web articles by the search country and the website owner entity type. The web article titled "Heart attack: First aid" written by Mayo Clinic Staff[28] was referenced in all chatbot responses generated for all studied countries, seemingly representing the main source of information. Similar to the chatbot responses, this web article omitted some guidelines-concordant instructions on first aid (e.g. to encourage the person to refrain from physical activity, get the person into a comfortable position, monitor person's breathing and

Table 3. Distribution of the source web articles referenced in the chatbot-generated responses

Website categories	The Gambia (*N* = 34), entity name, % (*n*)	India (*N* = 30), entity name, % (*n*)	The USA (*N* = 33), entity name, % (*n*)
Academic	Mayo Clinic, 67.6 (23)	Mayo Clinic, 66.7 (20)	Mayo Clinic, 60.6 (20)
Charity/ Nonprofit entity	n/a, 0.0 (0)	n/a, 0.0 (0)	St. Luke's Health, 3.0 (1)
Commercial	Healthline, 26.5 (9) The Healthy, 2.9 (1)	American Heart Association, 3.3 (1) TheHealthSite.com, 6.7 (2)	TheHealthSite.com, 3.0 (1)
Government	National Health Service, 2.9 (1)	National Health Service, 6.7 (2)	Centers for Disease Control and Prevention, 33.3 (11)
Social media	n/a, 0.0 (0)	HealthyHeartWorld.com, 3.3 (1) wikiHow, 13.3 (4)	n/a, 0.0 (0)

Abbreviation: n/a, not applicable.

responsiveness, bring an AED if possible if EMS is delayed) and contained inaccurate advice (unspecified dose of acetylsalicylic acid, inadequate criteria for starting CPR and using an AED, and admission of the possibility for the patient to drive themselves to a hospital in absence of other way to get help).

Of the unique source web articles (n = 11), on average 18.2% provided first aid instructions in full compliance with the checklist criteria, and 20.7% — in partial compliance (Appendix Table A). Eight (72.7%) of the unique web articles suggested superfluous actions contradictory to the guidelines (see Dataset[26]). Two (18.2%) of the articles were dedicated to clinical diagnosis and medical treatment of heart attack, but did not address first aid issues in any way.

Discussion

The present study was designed to test an ability of the novel LLM-powered chatbot, called the new Bing, to provide guidance on first aid in heart attack. The results show that the chatbot correctly understands the user query and generates moderately readable responses containing an advice on how to act in case of the emergency. Although the chatbot responses included a number of reasonable instructions concordant with current guidelines on first aid, the overall quality of the chatbot guidance was low. Firstly, the responses always omitted some important directions on first aid (for example, to encourage the person to refrain from physical

activity, to help the person to take their antianginal medication, or to start CPR if the person becomes unresponsive and stops breathing normally), which if followed may contribute to better outcomes in real cases of acute myocardial ischemia. Secondly, when relevant instructions were given, these were commonly inaccurate or incomplete, as exemplified by the suggestion of wrong EMS number, the advice to chew acetylsalicylic acid without specifying the recommended dose, or the indication of inadequate criteria for starting CPR or using an AED. Thirdly, a number of the chatbot responses contained excessive guidelines-conflicting directives which in case of a real emergency may distract the bystander's attention from the life-saving actions, or even cause additional harm to the person who needs help. Moreover, the content of the chatbot responses was nonconstant as for the number and accuracy of the instructions; these notably varied from one response to another.

Results of evaluation of the web articles utilised by the chatbot as a source for generating its responses suggest that the poor quality of the chatbot advice is largely determined by the source content. Inaccuracies and omissions from the web articles were carried by the chatbot to its responses along with the correct instructions on first aid.

With respect to the secondary objective, to compare the chatbot advice generated for different search geographies, it was found that for the studied countries the new Bing provided responses of variable but generally low quality, and did not adapt them to the search region. For instance, for the Gambia and India the chatbot failed to recommend the country-specific EMS numbers, and occasionally suggested use of an AED, while, to the best of our knowledge, there is no evidence of active public access defibrillation programs in these countries.

Prior to this research, several studies have tested an ability of the LLM-based chatbots to respond to user inquiries concerning help in health emergencies. J.E. Dahdah and colleagues (2023)[14] prompted ChatGPT with 30 "what to do" questions on different emergency conditions. Based on the assessment by emergency medicine specialists, 90% (n = 27) of the chatbot responses were graded as "appropriate" (including a response to the question "What to do if someone has chest pain?"), 10% (n = 3) −as "inappropriate," and no responses were rated as "harmful." I. Altamimi et al.[12] tested capabilities of ChatGPT to generate responses to questions related to acute venomous snakebites, including a question on first aid. According to the expert opinion provided by clinical toxicologists and emergency medicine physicians, the chatbot responses were accurate and informative. Less encouraging evidence as for the quality of the chatbot-generated advice on first aid has been received in our recent

study with the new Bing that evaluated an ability of the chatbot to provide instructions on help in foreign body airway obstruction.[13] That study showed that the chatbot commonly omitted potentially life-saving instructions on first aid, and generated inaccurate instructions not fully compliant with relevant first aid guidelines.

The current study adds to the body of exploratory research on the LLM chatbots' performance, and further calls reliability of the chatbot advice into question. Although the new Bing is powered by the next-generation OpenAI's language model GPT-4 that was declared by the developer to be more reliable compared to GPT-3.5 (the LLM of ChatGPT),[21] and despite it utilises complex machine learning algorithms to rank content found online and ground its responses on the most comprehensive, relevant, and valuable web search results,[20] for now the chatbot cannot be recommended as a credible source of information on first aid owing to the unacceptable quality of the chatbot-generated responses. Apparently, the new Bing's approach to the selection of source information based on the web content ranking is fundamentally inappropriate for answering public questions concerning health emergencies, as it is known that information on principles and techniques of bystander response in emergency conditions available online, even when published by widely recognized organizations and declared as guidelines-compliant, often has quality faults, which may not be obvious to nonexperts.[29-32]

Without doubt, any advice on health-related actions must be hundred-percent precise, up-to-date, complete and devoid of nonevidence-based suggestions. Therefore, before the publicly available chatbots will be improved to guarantee the highest possible quality of the AI-generated advice, developers of these tools should take prompt measures to mitigate risks of providing inaccurate and potentially harmful information, namely program the chatbots to refuse user requests for health-related advice as it was done for inquiries on how synthesize dangerous chemicals or how to create a bomb.[21] However, considering the potential of high-quality chatbot guidance to enhance rates of first aid provision and improve outcomes for life-threatening conditions, including cardiovascular emergencies, the chatbot developers, professional medical bodies and other stakeholders should seek for collaboration to create a comprehensive methodological framework for health-related chatbot counseling services, including a standardized expert-developed dataset of instructions on first aid that should be utilized as the sole and mandatory source of the advice. Wide public availability and steadily rising popularity of the LLM chatbots underscore the urgent necessity of the organizational interventions and continued research on the issue.

Study Limitations

The current study has limitations. It was an exploratory research that tested a particular LLM-based chatbot at a certain point in time. Although being made available for public in the preview mode, the new Bing chatbot is still under development. Other LLM-powered chatbots and the new Bing at a later stage of development may demonstrate different performance in terms of the responses' quality. For instance, for the new Bing it could be a consequence of modifications made to the web search results ranking algorithms. Changes of the web content itself would also probably influence the chatbot responses. Relatively small sample of responses included into the analysis should be considered as another limitation. Further, the study did not utilize Virtual Private Network because of unavailability of such service for the Gambia. Although the countries' capitals were set as a search location in the Bing settings, absence of the Virtual Private Network may have limited our ability to simulate search from a specific city's or country's network address.

Conclusions

The chatbot-generated responses to user inquiries concerning provision of help in heart attack have omissions, inaccuracies and misleading instructions which are carried forward from the web sources. Public availability of the chatbot and its readiness to give plausible-sounding, yet unreliable, advice on first aid in a health emergency create the risk of uncontrolled misinformation and possible harm. Relevant stakeholders should prioritize collaborative efforts aimed at ensuring the trustworthiness, standardization and proper human oversight of the chatbot-mediated counseling. Active research efforts are necessary to better understand potential implications of the AI-driven informational support in health crises and to work out effective strategies for mitigating associated risks.

Research Data for this Article

All original data generated in this research are publicly available in online repository as a dataset.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix Table A. Congruence of the unique source web articles' (*N* = 11) content with first aid guidelines as judged by satisfaction of the checklist criteria

Checklist criteria	Completely satisfied, % (*n*)	Partially satisfied, % (*n*)	Completely or partially satisfied, % (*n*)
1. Does the response instruct to help the person to get into a comfortable position?	9.1 (1)	18.2 (2)	27.3 (3)
2. Does the response instruct to encourage the person to refrain from physical activity?	18.2 (2)	9.1 (1)	27.3 (3)
3. Does the response instruct to help the person to take their antianginal medication?	36.4 (4)	9.1 (1)	45.5 (5)
4. Does the response instruct to access EMS immediately?	63.6 (7)	9.1 (1)	72.7 (8)
5. Does the response instruct to ask a bystander to bring an AED if possible if EMS is delayed?	0.0 (0)	27.3 (3)	27.3 (3)
6. Does the response instruct to suggest the person to consider chewing a dose of acetylsalicylic acid (150-300 mg)?	27.3 (3)	45.5 (5)	72.7 (8)
7. Does the response instruct not to give acetylsalicylic acid to the person if they are allergic to it, have a bleeding disorder, or have already taken the recommended dose?	0.0 (0)	45.5 (5)	45.5 (5)
8. Does the response instruct to reassure the person?	27.3 (3)	0.0 (0)	27.3 (3)
9. Does the response instruct to monitor person's breathing and responsiveness?	9.1 (1)	9.1 (1)	18.2 (2)
10. Does the response instruct to start CPR if the person becomes unresponsive with abnormal breathing?	0.0 (0)	27.3 (3)	27.3 (3)
11. Does the response instruct to use an AED if available if the person becomes unresponsive with abnormal breathing?	9.1 (1)	27.3 (3)	36.4 (4)
Mean percentage	18.2	20.7	38.9

Abbreviations: AED, automated external defibrillator; CPR, cardiopulmonary resuscitation; EMS, emergency medical services.

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